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EVENT REPORT ON

"Manufacturing of Flange Coupling"

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1. INTRODUCTION

In modern manufacturing industries, accuracy, productivity, and repeatability are essential requirements for producing mechanical components. With the rapid development of technology, traditional machining methods are increasingly being replaced by advanced automated systems. One such important advancement is **Computer Numerical Control (CNC)** machining, which has revolutionized the manufacturing process by enabling precise and efficient production of complex components.

CNC machining is a manufacturing process in which machine tools are controlled automatically by a computer using numerical instructions. These instructions are programmed in the form of codes, commonly known as **G-codes and M-codes**, which direct the movement of cutting tools and control machining operations such as turning, milling, drilling, and boring. CNC machines reduce the dependence on manual operation, thereby minimizing human error and improving consistency in product quality.

A **flange coupling** is a rigid mechanical component used in power transmission systems to connect two co-axial shafts and transmit torque from one shaft to another. It consists of two circular flanges mounted on the ends of shafts and bolted together to form a rigid connection. Flange couplings are widely used in industrial machinery such as pumps, compressors, generators, and turbines due to their simplicity, strength, and reliability.

Manufacturing a flange coupling requires high dimensional accuracy, proper alignment, and good surface finish to ensure efficient torque transmission and long service life. CNC machines are ideally suited for this purpose, as they can produce flange couplings with precise dimensions, accurate bore sizes, and perfectly aligned bolt holes. Operations such as facing, turning, boring, drilling, and tapping can be carried out efficiently using CNC turning and CNC milling machines.

This project focuses on the **manufacturing of an unprotected flange coupling using CNC machines**. The study includes an overview of CNC technology, design considerations, machining operations, and advantages of CNC manufacturing. The use of CNC machines in flange coupling production ensures improved productivity, consistent quality, and suitability for modern industrial applications.

2. OVERVIEW OF CNC TECHNOLOGY

Computer Numerical Control (CNC) is a manufacturing process in which machine tools are controlled by a computer executing a pre-written program. CNC machines operate based on numerical data, coordinates, and instructions known as G-codes and M-codes.

Unlike conventional machines, CNC machines eliminate human error, ensure consistent product quality, and allow mass production of components with minimal variation.

Key advantages of CNC technology include high accuracy, repeatability, reduced production time, improved safety, and the ability to manufacture complex geometries.



CNC (Computer Numerical Control) is a manufacturing system in which machine tools are operated automatically by a computer using programmed numerical instructions. These instructions, called **G-codes and M-codes**, control the movement of the cutting tool and workpiece to perform machining operations such as turning, milling, drilling, and boring.

In conventional machines, operations are performed manually by the operator, which can lead to errors and variation in accuracy. CNC machines overcome these

limitations by providing **high precision, repeatability, and consistent quality**. Once a program is prepared and loaded into the CNC machine, identical components can be produced continuously without human intervention.

The working of a CNC machine starts with the creation of a component design using **CAD (Computer-Aided Design)** software. This design is then converted into machining instructions using **CAM (Computer-Aided Manufacturing)** software. The generated CNC program is transferred to the machine, where the **Machine Control Unit (MCU)** interprets the code and controls spindle speed, feed rate, tool movement, and cutting operations.

The main parts of a CNC machine include the **Machine Control Unit (MCU)**, drive system, machine tool, and work-holding device. The MCU acts as the brain of the machine and controls all operations. Servo motors and ball screws provide accurate movement along the X, Y, and Z axes. Chucks, vices, or fixtures are used to hold the workpiece firmly during machining.

CNC machines are classified into CNC lathe machines, CNC milling machines, CNC machining centers, and CNC drilling machines. CNC lathe machines are mainly used for facing, turning, and boring operations, while CNC milling machines are used for milling, drilling, tapping, and slotting operations.

CNC machines offer several advantages such as high dimensional accuracy, excellent surface finish, reduced production time, low human error, and suitability for mass production. However, they also have some limitations, including high initial cost and the requirement of skilled operators.

CNC technology is widely used in automotive, aerospace, medical, and manufacturing industries. In mechanical engineering, CNC machines are commonly used for producing precision components such as **shafts, gears, and flange couplings**.

3. FLANGE COUPLING

A **flange coupling** is a rigid mechanical device used to connect two co-axial shafts for the purpose of transmitting torque and rotational motion from one shaft to another. It is one of the most commonly used couplings in mechanical power transmission systems due to its simple construction, high strength, and reliability. Flange couplings are mainly used in applications where shafts are perfectly aligned and where high torque transmission is required.

In industries, flange couplings are widely used in pumps, compressors, generators, turbines, and heavy machinery. Their rigid nature ensures minimal power loss during transmission and provides a strong and permanent shaft connection.

a) Construction of Flange Coupling

A flange coupling consists of the following main parts:

1. Flanges

Two circular flanges are mounted on the ends of the shafts. Each flange has a central hole (bore) to fit onto the shaft.

2. Shaft

The shafts are the rotating members that transmit power.

3. Bolts and Nuts

The two flanges are connected together using bolts and nuts arranged on a pitch circle diameter (PCD).

4. Key and Keyway

A key may be provided between the shaft and flange to prevent slipping and ensure positive drive.

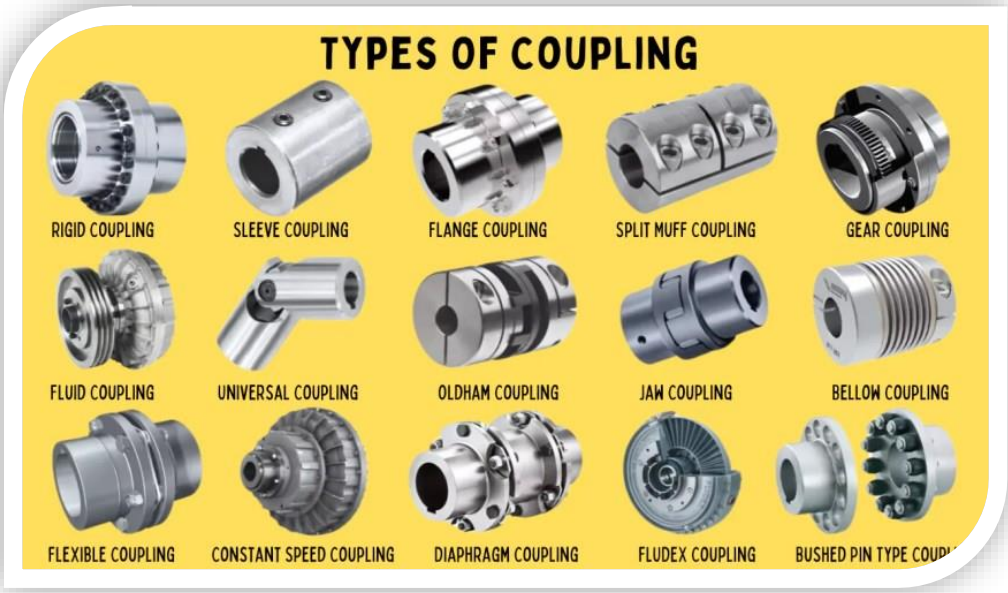
b) Working Principle

The working principle of a flange coupling is simple. When the driving shaft rotates, the flange mounted on it also rotates. The torque is transmitted to the second flange through the bolts joining the two flanges. From the second flange, the torque is transferred to the driven shaft.

c) Types of Flange Couplings

Flange couplings are classified into the following types:

1) Unprotected Flange Coupling
2) Protected Flange Coupling
3) Marine Type Flange Coupling



1) UNPROTECTED FLANGE COUPLING

An unprotected flange coupling is a rigid coupling used to connect two co-axial shafts for transmitting torque. It consists of two circular flanges mounted on the shaft ends and bolted together. In this type, the bolts and nuts are exposed, and no protective cover is provided.



Each flange is fixed to the shaft using a key and keyway arrangement. When the driving shaft rotates, torque is transmitted to the driven shaft through the bolts and friction between the flange faces. Since it is a rigid coupling, it does not permit any shaft misalignment.

Unprotected flange couplings are simple in construction, easy to inspect, and capable of transmitting high torque. They are commonly used in pumps, compressors, and heavy machinery where proper alignment and safety precautions are maintained

d) Advantages of Flange Coupling

- Simple and robust construction
- Capable of transmitting high torque
- High strength and durability
- Easy to assemble and dismantle
- Suitable for heavy-duty applications

e) Disadvantages of Flange Coupling

- Cannot accommodate shaft misalignment
- Rigid connection may cause vibration
- Requires accurate shaft alignment
- Not suitable for shock loads

f) Applications of Flange Coupling

Flange couplings are commonly used in:

- Pumps and compressors
- Power plants
- Industrial machinery
- Marine propulsion systems
- Turbines and generators

4. MATERIAL SELECTION FOR FLANGE COUPLING

Material selection plays a crucial role in the performance, strength, durability, and service life of a flange coupling. Since a flange coupling is used to transmit torque between two co-axial shafts, the selected material must be capable of withstanding high torsional loads, stresses, vibration, and wear during operation. Proper material selection ensures safe operation, long life, and reliable power transmission.

- High tensile and shear strength
- Good toughness and fatigue resistance
- Adequate machinability
- Resistance to wear and corrosion (where required)
- Cost effectiveness and easy availability

Common Materials Used for Flange Couplings

1. **Mild Steel (MS)**

   Mild steel is the most commonly used material for manufacturing flange couplings. It has good strength, excellent machinability, and is economical. Mild steel can withstand moderate to high torque and is easy to machine using conventional and CNC machines. Due to its balanced mechanical properties and low cost, it is widely preferred in general industrial applications.

2. **Alloy Steel**

   Alloy steels are used in heavy-duty and high-torque applications. They provide higher strength, better wear resistance, and improved fatigue life compared to mild steel. However, alloy steels are more expensive and difficult to machine, so they are used only when high performance is required.

3. **Stainless Steel (Special Applications)**

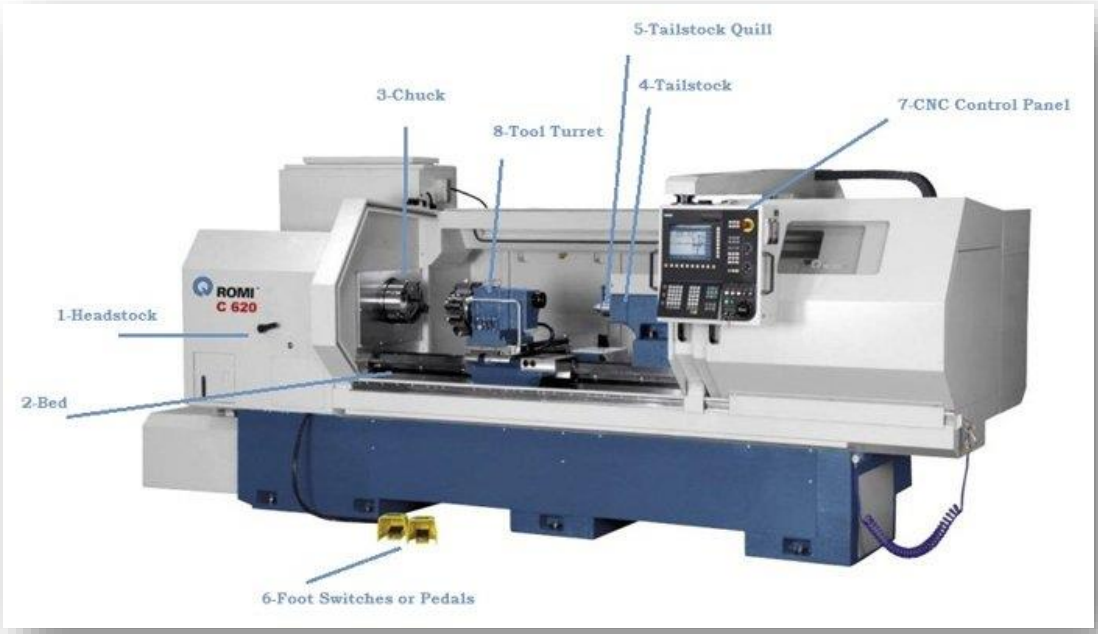
   Stainless steel is used in corrosive environments such as chemical industries or marine applications. It offers good corrosion resistance but is costly and

5. CNC MACHINES USED

In the manufacturing of a flange coupling, **CNC machines** are used to achieve high accuracy, good surface finish, and repeatability. The main CNC machines used are the **CNC Lathe Machine** and the **CNC Milling Machine** Each machine performs specific operations required to manufacture the flange coupling as per design specifications.

1. CNC Lathe Machine

The CNC lathe machine is primarily used for machining rotational components. In flange coupling manufacturing, the CNC lathe performs most of the basic shaping operations. The workpiece is held in a chuck and rotates at a specified spindle speed, while the cutting tool moves along programmed paths.



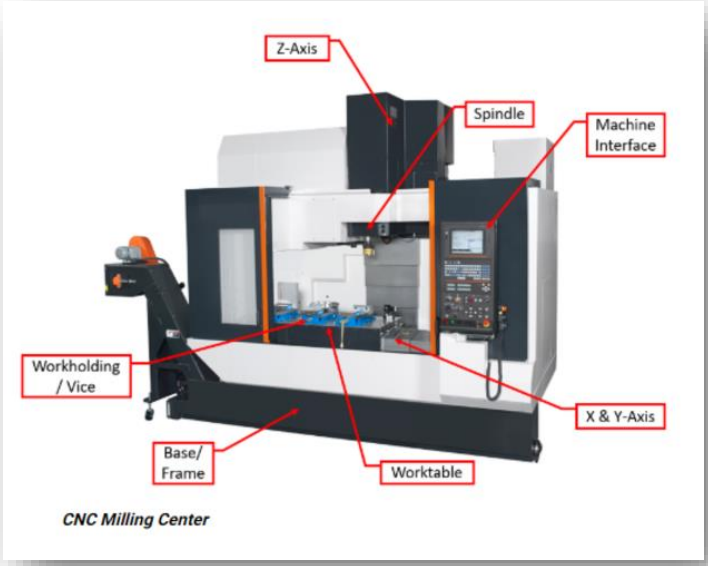
Operations performed on the CNC lathe include:

- Facing to obtain a flat surface
- Rough turning to remove excess material
- Finish turning to achieve final diameter and surface finish
- Boring to produce the central hole for shaft mounting

The CNC lathe ensures accurate control of dimensions such as outer diameter, flange thickness, and bore size. Programs are written using G-codes such as G01 for linear cutting, G71 for rough turning, and G70 for finish turning.

2. CNC Milling Machine

The CNC milling machine is used for machining features that cannot be produced on a lathe. In flange coupling manufacturing, it is mainly used for hole-making operations. The workpiece is clamped on the worktable, and the rotating



CNC Milling Center

cutting tool performs machining.

Operations performed on the CNC milling machine include:

- Drilling bolt holes on the pitch circle diameter
- Counterboring for bolt heads
- Tapping for bolt threads
- Milling keyways

6. CAD DESIGN OF FLANGE COUPLING

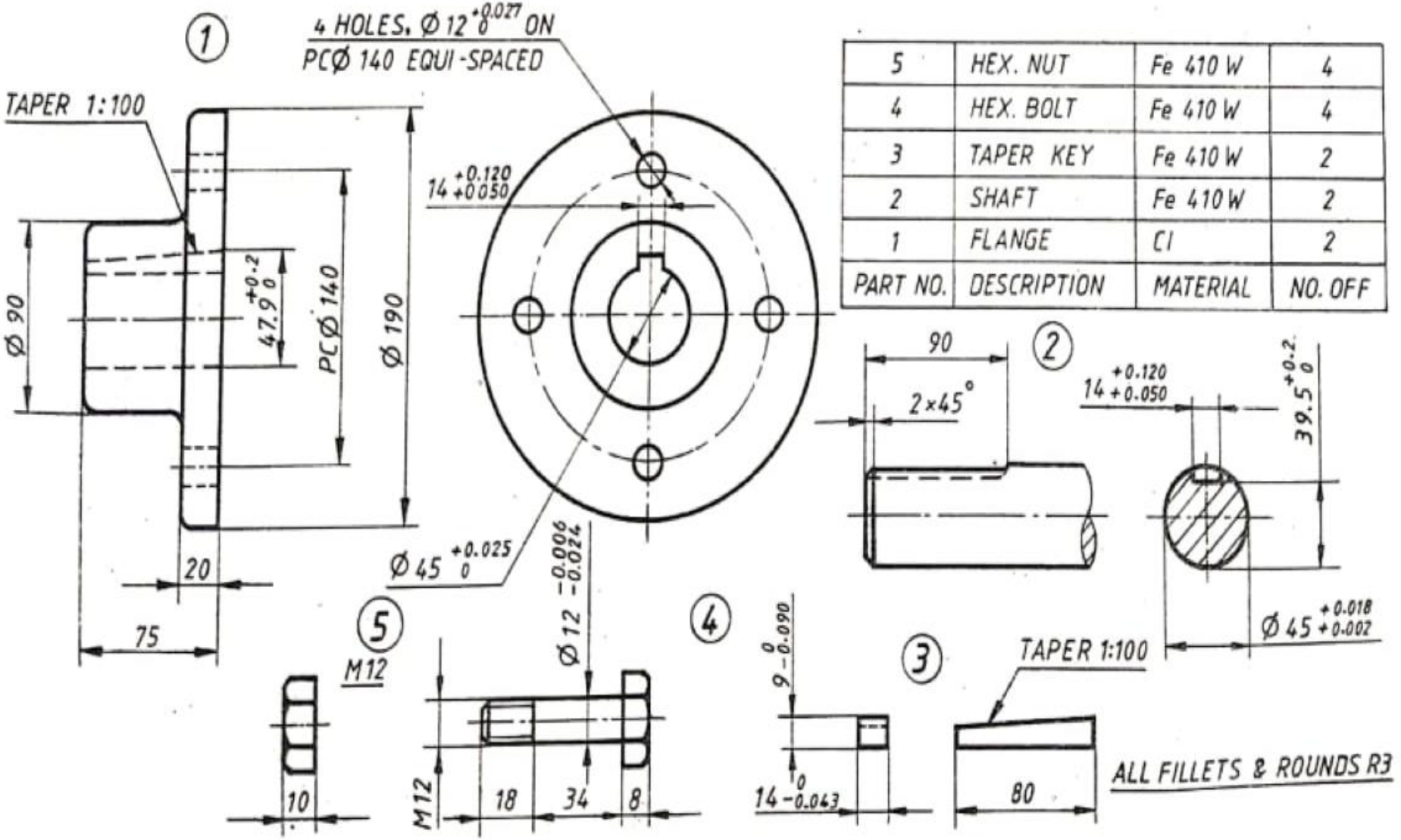
Computer-Aided Design (CAD) plays an important role in the accurate and efficient design of mechanical components. In the manufacturing of a flange coupling, CAD software is used to create detailed **2D drawings** and **3D models** that define the exact shape, size, and features of the coupling. CAD design helps in visualizing the component before manufacturing and ensures that all dimensions meet functional and assembly requirements.

The CAD design of a flange coupling begins with selecting the type of coupling, usually an **unprotected flange coupling**, based on application requirements. The main dimensions such as shaft diameter, flange outer diameter, flange thickness, bolt circle diameter, number of bolt holes, and bolt size are decided using standard design equations and mechanical design data.

A **2D drawing** of the flange coupling is created first. This drawing includes all critical dimensions, tolerances, and annotations. It shows the front view, top view, and sectional view of the flange. The central bore for shaft mounting, keyway details, and bolt hole positions on the pitch circle diameter are clearly indicated. The 2D drawing is essential for manufacturing and inspection purposes.

After completing the 2D drawing, a **3D model** of the flange coupling is developed using CAD software such as SolidWorks or AutoCAD. The 3D model provides a clear visualization of the actual component shape and helps in checking assembly fit and interference. It also helps in understanding how the two flanges align and bolt together during assembly.

The CAD model ensures dimensional accuracy and acts as the base input for **CAM (Computer-Aided Manufacturing)** software. The CAD file is directly imported into CAM software to generate tool paths and CNC programs for machining operations such as turning, boring, drilling, and tapping.



PART NO.	DESCRIPTION	MATERIAL	NO. OFF
5	HEX. NUT	Fe 410 W	4
4	HEX. BOLT	Fe 410 W	4
3	TAPER KEY	Fe 410 W	2
2	SHAFT	Fe 410 W	2
1	FLANGE	CI	2

All Dimensions in mm

Details of Flanged Coupling — Unprotected Type

7. CAM AND CNC PROGRAMMING

The CAD model is imported into CAM software, where toolpaths are generated. Based on these toolpaths, CNC programs are created using G-codes and M-codes.

CNC Turning Program

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N00 G21 G90 G98 G40;
N10 G28 U0W0;
N20 T0101;
N30 G50 S1000 M03;
N40 G96 S100 M08;
N50 G00 X87 Z0;
N60 G71 U0.5 R1;
N70 G71 P80 Q140 U0 W0 F2;
N80 G01 X37;
N90 G03 X40 Z-3 R3;
N100 G01 Z-37;
N110 G02 X43 Z-40 R3;
N120 G01 X82;
N130 G03 X85 Z-43 R3;
N140 G01 Z-65;
N150 G70 P80 Q140 F100;
N160 G28 U0W0 ;
N170 M05;
N180 M09;
N190 M30;
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Rough turning is performed using the G71 cycle, while finishing is done using the G70 cycle. Spindle speed, feed rate, and depth of cut are selected based on material and tool.

8. MACHINING OPERATIONS

Machining operations are the most important stage in the manufacturing of a flange coupling. These operations are carried out using **CNC turning and CNC milling machines** to achieve the required shape, dimensions, and surface finish. Proper sequence of machining operations ensures dimensional accuracy, good surface quality, and correct assembly of the coupling.

8.1 Facing Operation

Facing is the first operation performed on the CNC lathe. In this operation, the cutting tool moves perpendicular to the axis of rotation to produce a flat and smooth surface on the workpiece. Facing ensures that the end surface of the flange is flat and perpendicular to the shaft axis, which is essential for proper alignment and assembly.

8.2 Rough Turning Operation

Rough turning is carried out to remove excess material from the workpiece quickly. In this operation, a higher depth of cut and feed rate are used. The outer diameter of the flange is reduced close to the required size. Rough turning improves productivity and prepares the workpiece for finishing operations. In CNC machines, this operation is commonly performed using the **G71 rough turning cycle**.

8.3 Finish Turning Operation

Finish turning is performed after rough turning to achieve the final dimensions and required surface finish. A smaller depth of cut and lower feed rate are used in this operation. Finish turning ensures accurate outer diameter, flange thickness, and smooth surface quality. The **G70 finishing cycle** is generally used in CNC turning.

8.4 Boring Operation

Boring is performed to machine the central hole of the flange that fits onto the shaft. This operation ensures accurate bore diameter, proper concentricity, and correct fit between the shaft and the flange. Boring is a critical operation because improper bore size can lead to vibration or slipping during operation.

8.5 Drilling Operation

Drilling is carried out on a CNC milling machine to produce bolt holes on the pitch circle diameter of the flange. Accurate positioning of bolt holes is essential for proper alignment of the two flanges during assembly. CNC milling ensures precise hole location and uniform spacing.

9. INSPECTION AND QUALITY CONTROL

Inspection and quality control are essential steps in the manufacturing of a flange coupling to ensure that the final product meets design specifications, dimensional accuracy, and functional requirements. Proper inspection helps in detecting manufacturing defects, reducing rework, and improving product reliability and service life.

After machining operations are completed, the flange coupling is subjected to various inspection procedures. The **outer diameter**, **flange thickness**, and **overall dimensions** are measured using instruments such as vernier calipers and micrometers. These measurements ensure that the component conforms to the specified tolerances mentioned in the design drawing.

The **bore diameter** of the flange, which fits onto the shaft, is a critical dimension and is inspected using inside micrometers or plug gauges. Accurate bore size ensures proper fit, prevents slippage, and avoids vibration during operation. Concentricity between the bore and outer diameter is also checked to ensure smooth rotation.

Bolt holes drilled on the pitch circle diameter are inspected for correct position, size, and spacing. Vernier calipers, templates, or coordinate measurement methods are used to verify bolt hole alignment. Proper alignment of bolt holes is necessary for easy assembly of the two flanges.

Surface finish inspection is carried out visually or using surface roughness comparison standards. A good surface finish ensures proper contact between mating flange faces and improves torque transmission. Threads produced by tapping operations are checked using thread gauges to ensure correct thread size and pitch.

10. SAFETY PRECAUTIONS

Safety precautions are very important during the manufacturing of a flange coupling using CNC machines. Proper safety practices help in preventing accidents, protecting operators, and ensuring smooth machining operations. Since CNC machines involve high-speed rotating parts, sharp cutting tools, and electrical systems, strict safety rules must be followed.

Before starting the CNC machine, the operator should ensure that the workpiece is properly clamped in the chuck or fixture. Loose workpieces can cause serious accidents during machining. Cutting tools should be correctly mounted and checked for wear or damage. Only the required tool should be loaded in the tool turret.

The CNC program must be verified and simulated before actual machining. Program errors can cause tool collision, machine damage, or injury to the operator. Proper spindle speed, feed rate, and depth of cut should be selected according to the material and tool.

During machining, the safety door of the CNC machine must be closed. Operators should never touch rotating parts or cutting tools while the machine is running. Coolant flow should be properly maintained to avoid overheating of tools and workpieces.

Personal protective equipment such as safety goggles, gloves (when handling workpieces), and safety shoes should be worn. Loose clothing, jewelry, or long hair must be avoided near rotating machinery.

After machining, the machine should be stopped completely before removing the workpiece. Chips and swarf should be cleaned using a brush or air gun and never by hand. Electrical panels should be handled only by authorized personnel.

11. CONCLUSION

The manufacturing of a flange coupling using CNC machines demonstrates the importance of modern machining techniques in achieving high precision, accuracy, and productivity. CNC technology enables the production of flange couplings with consistent dimensions, good surface finish, and reliable quality, which are essential for efficient power transmission in mechanical systems.

In this work, the flange coupling was designed using CAD software and manufactured using CNC turning and CNC milling machines. Operations such as facing, rough turning, finish turning, boring, drilling, and tapping were carried out in a systematic manner. The use of CNC machines ensured proper control of machining parameters, reduced human error, and improved repeatability of the component.

Inspection and quality control procedures confirmed that the manufactured flange coupling met the required design specifications and tolerances. Safety precautions followed during machining helped in preventing accidents and ensured smooth operation of the machines.

Overall, CNC machining proves to be an effective and reliable method for manufacturing flange couplings used in industrial applications such as pumps, compressors, turbines, and power transmission systems. The study highlights that CNC-based manufacturing not only improves product quality but also increases production efficiency, making it suitable for modern manufacturing industries.